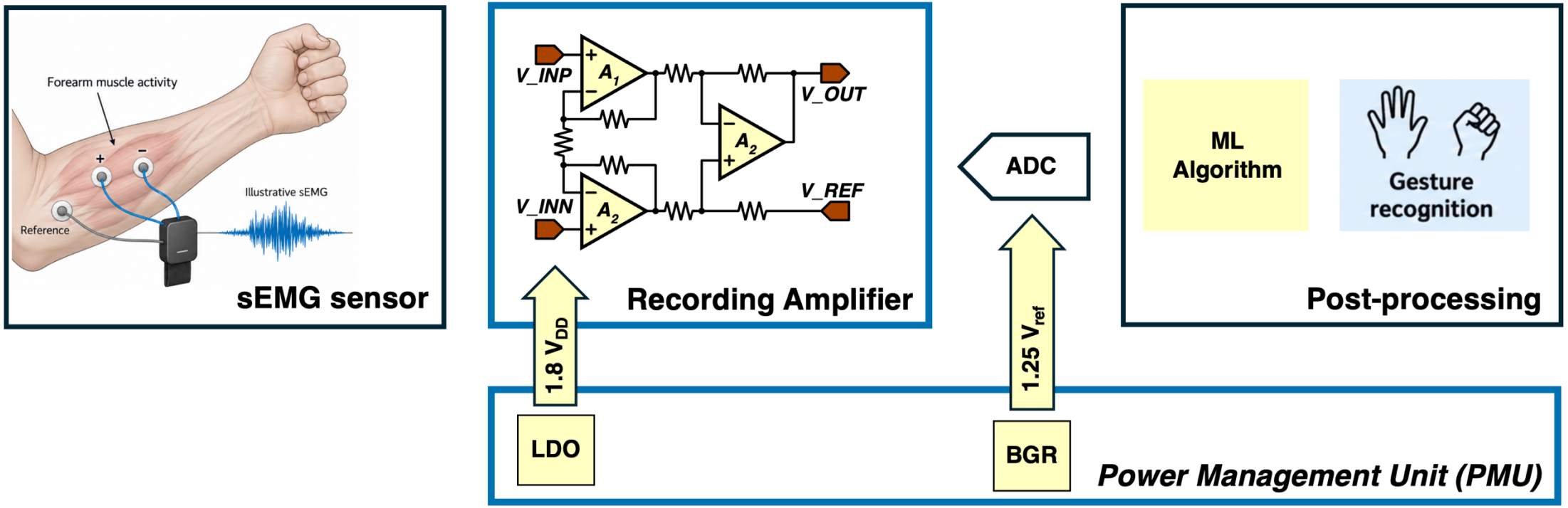
EEMEM0006_2026
TB-1
AIC coursework brief

Chen, Junzhe (JC)

02/Nov/2026, Bristol, UK

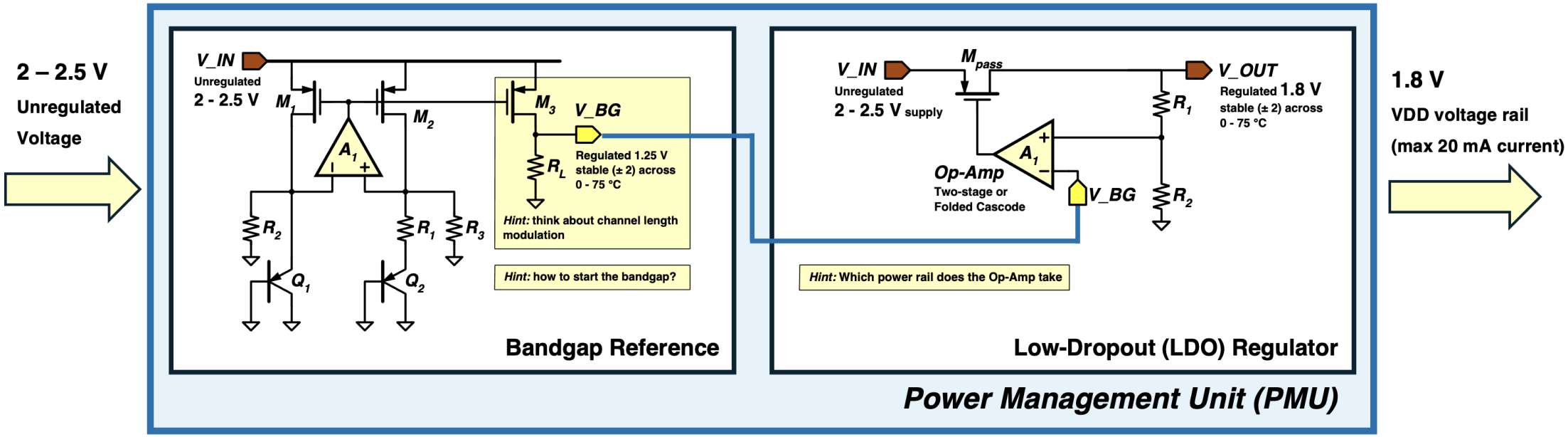
Introduction



Design parts highlighted in blue

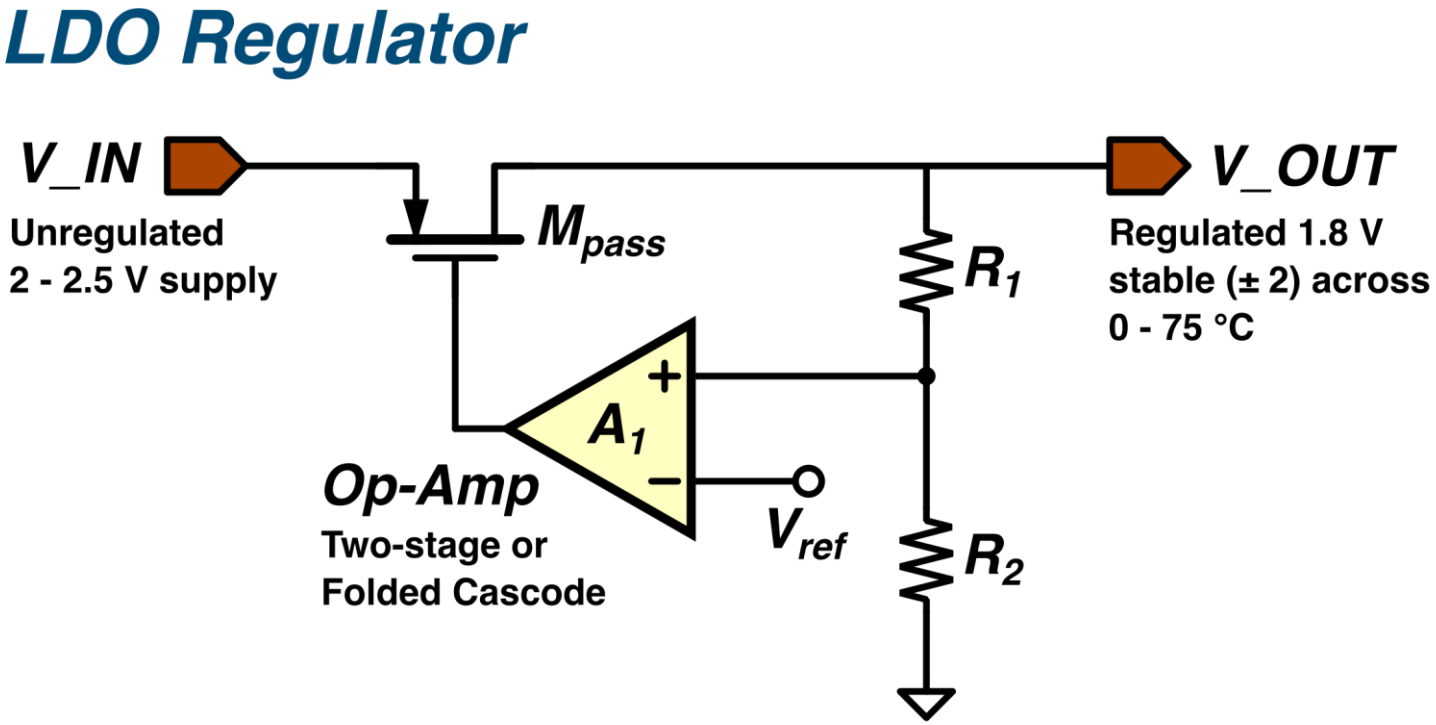
Power Management Unit (PMU)

- Input: 2 – 2.5 V unregulated voltage supply (e.g. a battery)
- Output: 1.8 V regulated voltage supply (for recording amplifier)



Low-Dropout (LDO) regulator

- Pass transistor (controlled resistor)
- Error Amplifier with Resistive Feedback

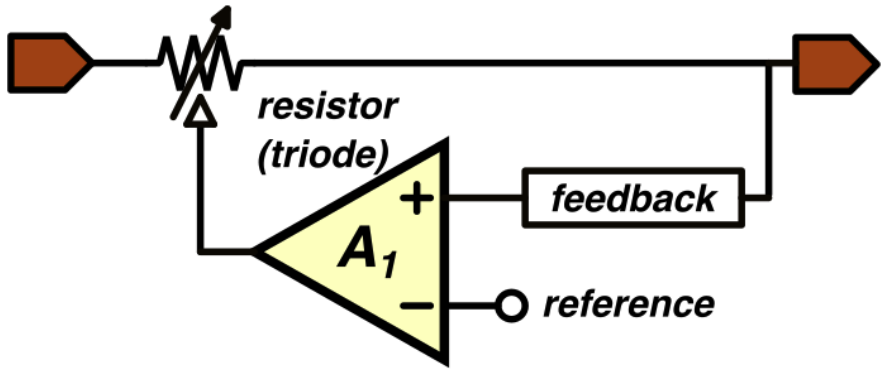


Low-Dropout (LDO) regulator

- Pass transistor (controlled resistor)
- Error Amplifier with Resistive Feedback

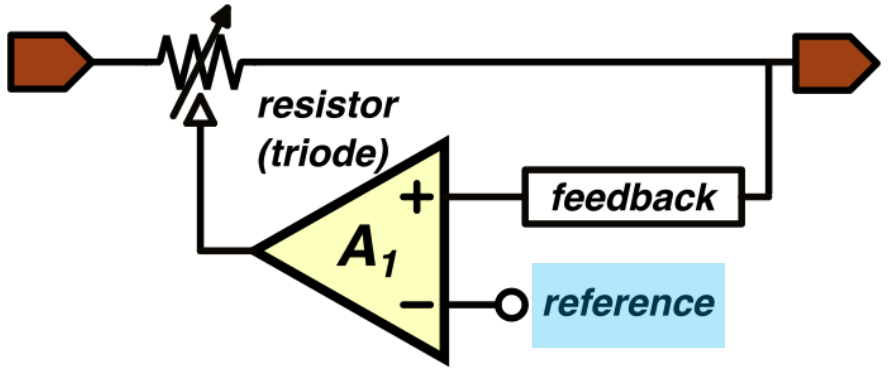
$$V_{in} - I_{load}R_{M_{pass}} = 1.8\,V$$

dropout



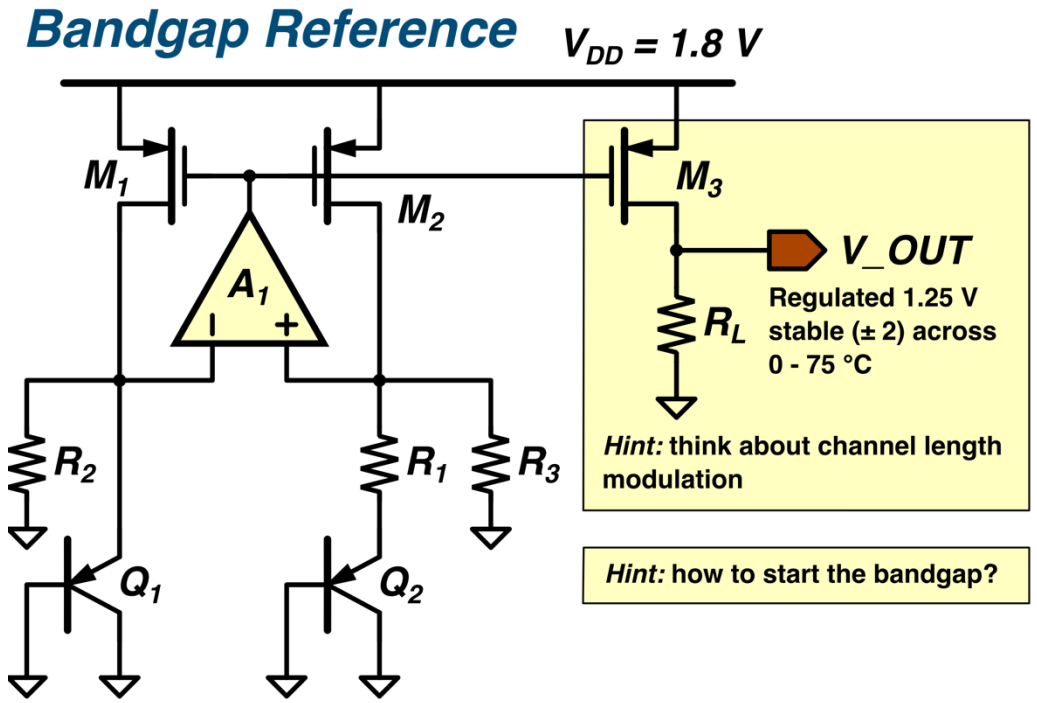
Low-Dropout (LDO) regulator

- Pass transistor (controlled resistor)
- Error Amplifier with Resistive Feedback
- Where to get the reference?



Bandgap Reference (BGR)

- Precise voltage reference
- Independent of temperature

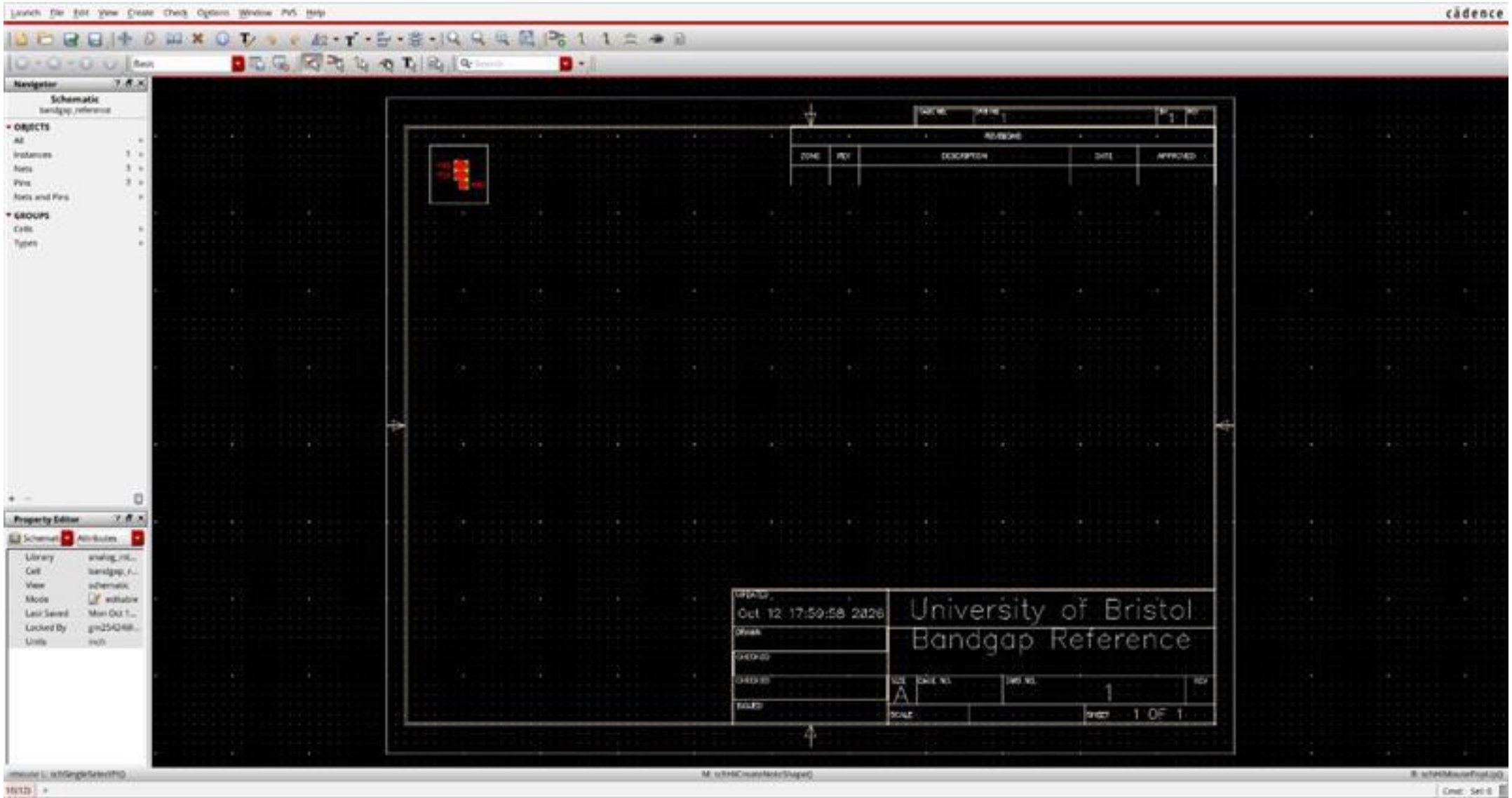


Start the coursework

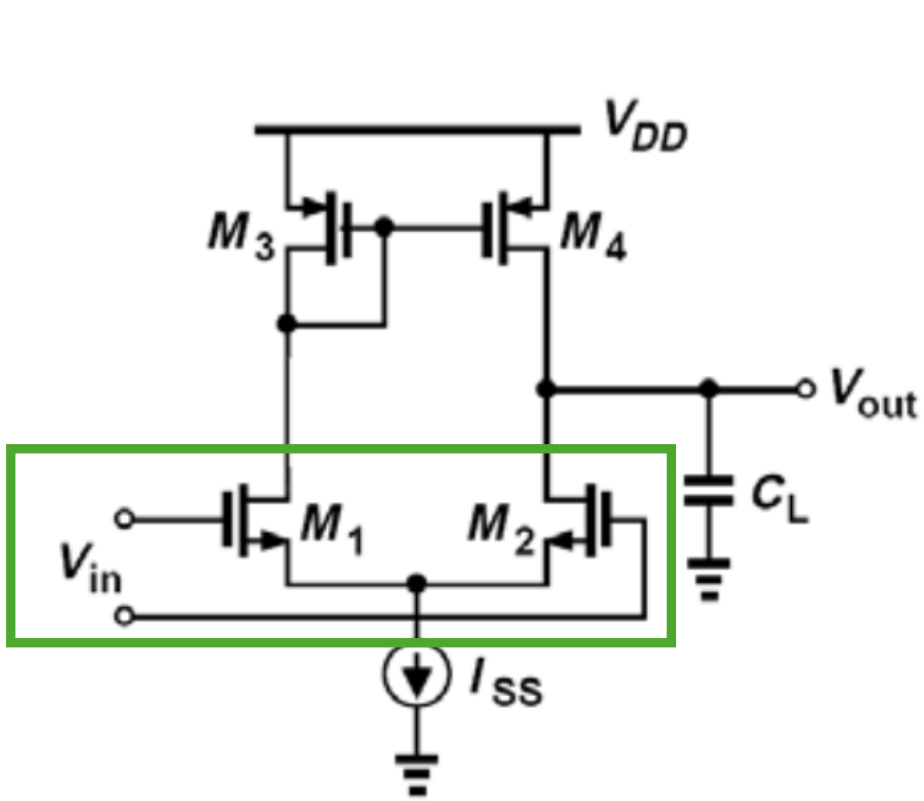
1. Clone the repo from GitHub
   https://github.com/Junzhe-Chen/Analogue_Integrated_Circuit_projects
2. Copy `cw_submission` to your working directory
3. Follow the chapter, clear all the specs
4. If you want to do it at home, see Appendix

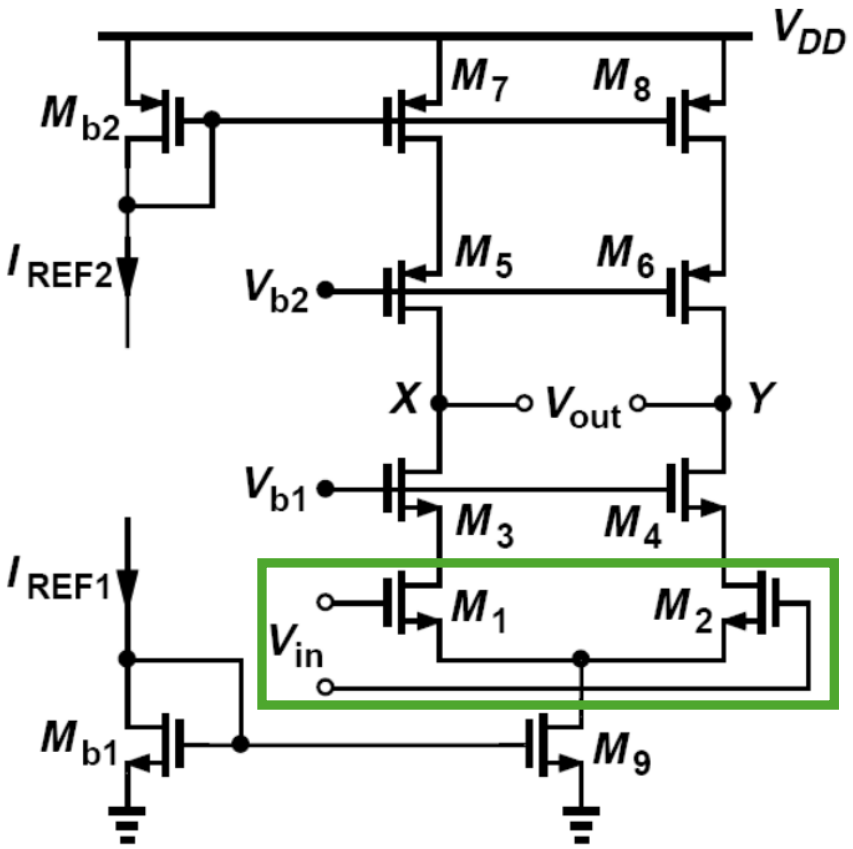
   Reminder: remote connection is *extremely* slow

Design sheets



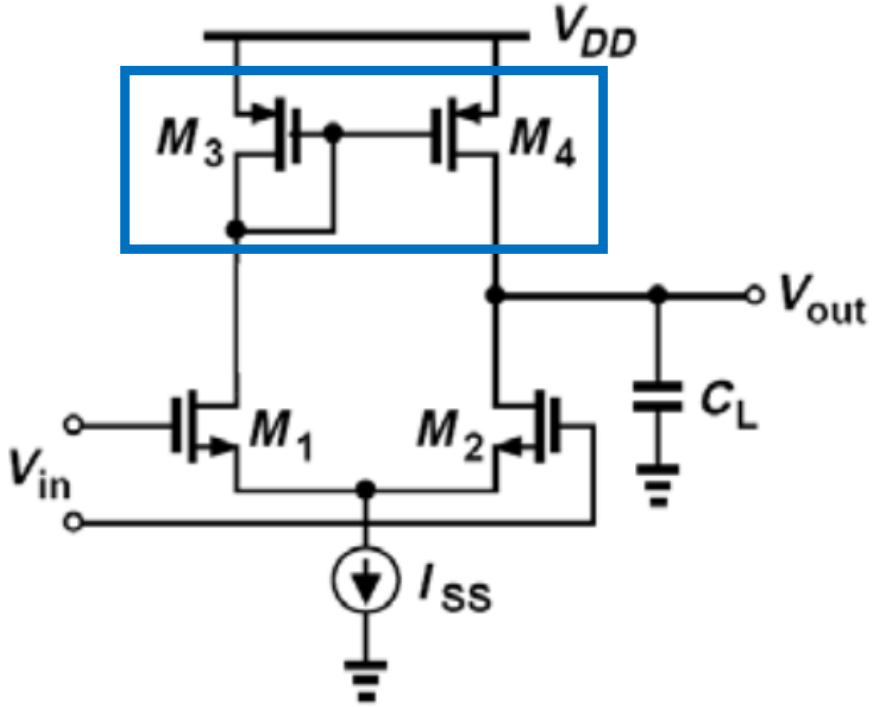
Design Trick: Read the Block

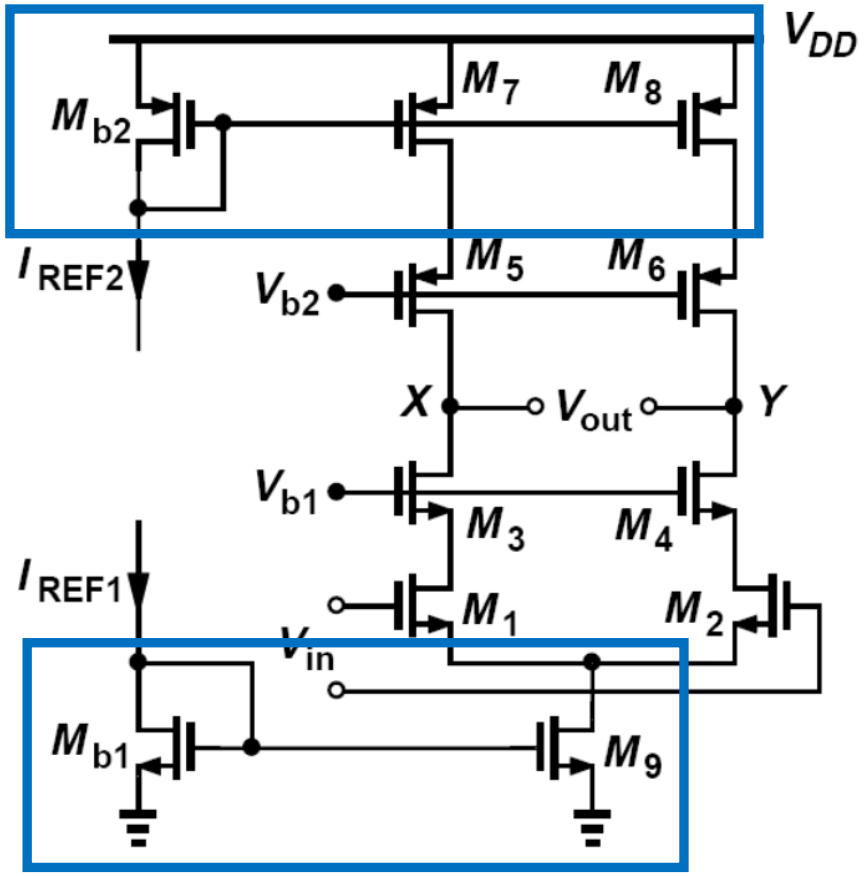




Differential pair (amplify differential signal)

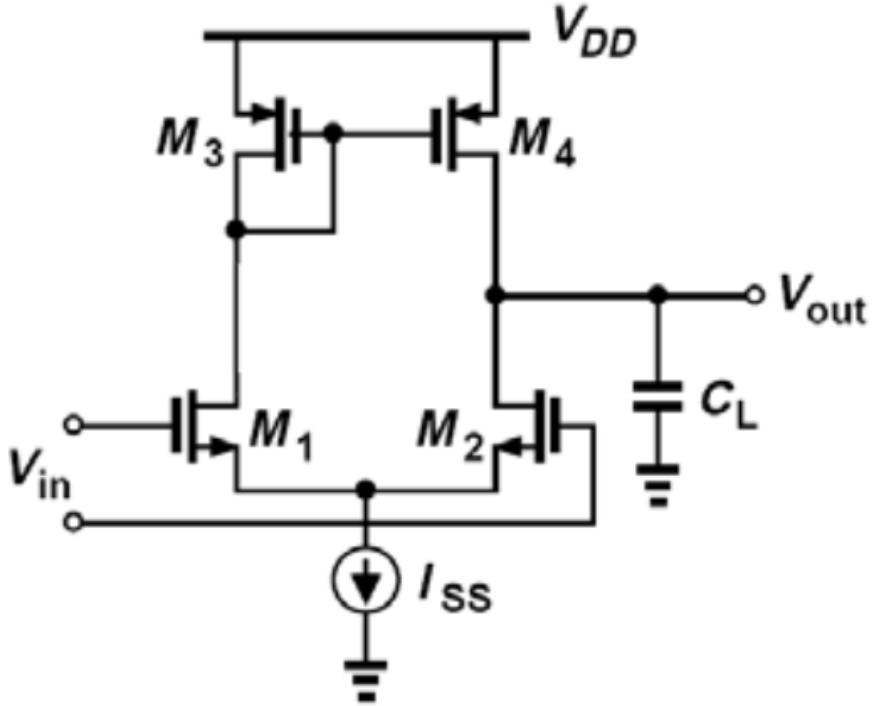
Design Trick: Read the Block

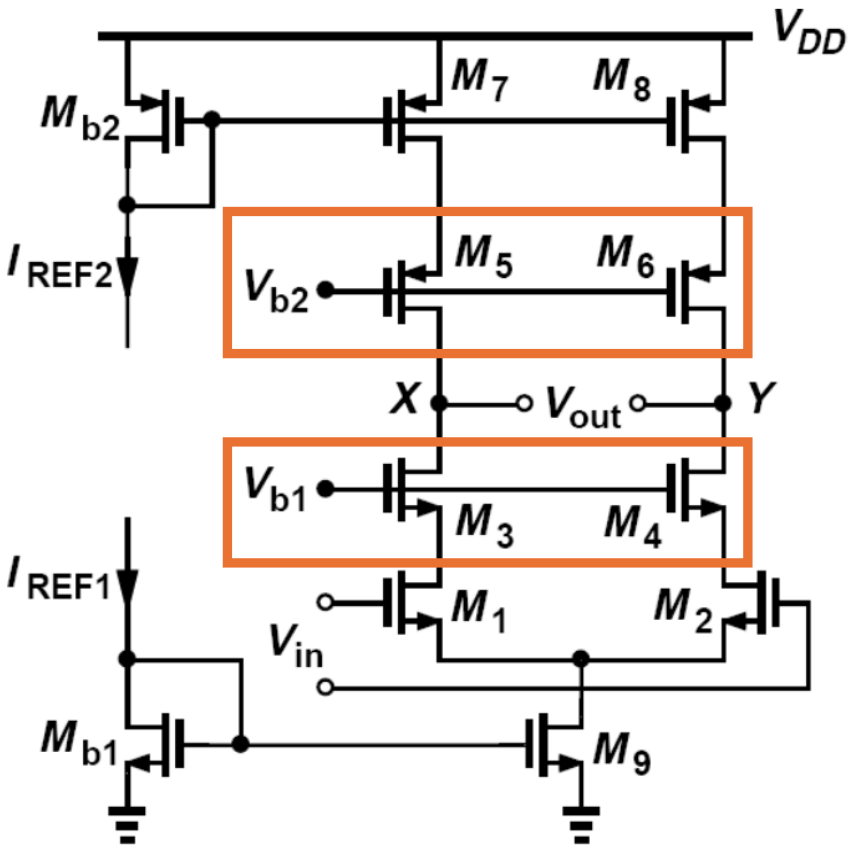




Current Mirror (copy currents)

Design Trick: Read the Block

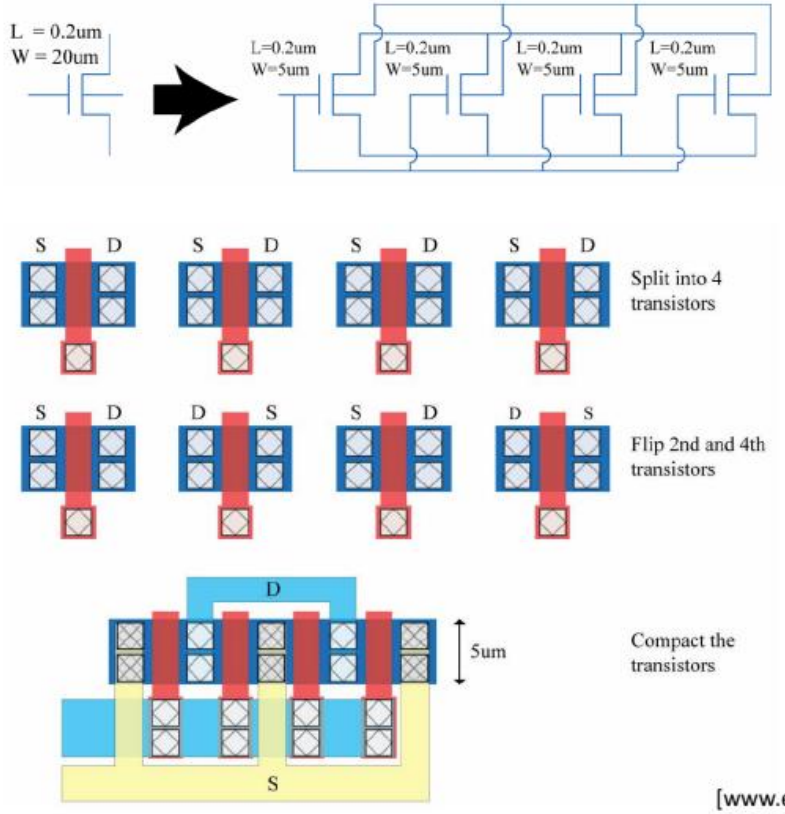




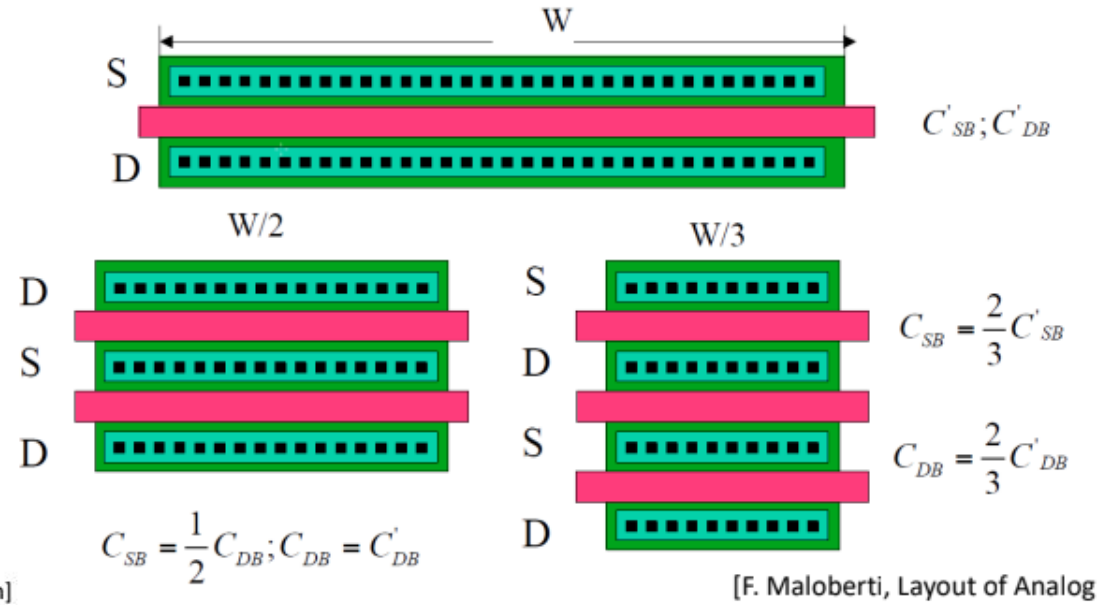
Cascode (gain improvement)

Design Trick: Fingers and Multipliers

- Break one long (large W) device into several smaller devices
- Smaller parasitic capacitors and gate resistance



[www.eda-utilities.com]



[F. Maloberti, Layout of Analog CMOS IC]